# RESEARCH PROJECT PROPOSAL

**Project Title: SPHERE** (Stakeholder Prioritization and Holistic Environmental Requirements Engine)

**Domain:** Natural Language Processing (NLP) / Machine Learning / Decision Support Systems

## 1. Abstract

During the planning phases of major infrastructure megaprojects, urban planners are legally required to conduct Environmental Impact Assessments (EIAs) and open public consultation periods. The transition to digital e-participation has created a massive data-processing bottleneck, with planners receiving tens of thousands of unstructured, qualitative comments from stakeholders.

This project proposes an automated, end-to-end Natural Language Processing (NLP) pipeline designed to extract, cluster, and prioritize sustainability requirements directly from unstructured citizen feedback. Utilizing a fine-tuned DistilBERT classifier, a custom Named Entity Recognition (NER) model, UMAP/HDBSCAN clustering, and a Large Language Model (LLM) prioritization cascade, the proposed system transforms noisy public chatter into a structured, actionable decision-support matrix for sustainable urban planning.

## 2. Introduction & Motivation

The integration of sustainable practices into civil engineering is a critical global priority. However, as noted by Falco et al. (2019), "massive digital participation" has overwhelmed traditional urban planning mechanisms. Because analysts lack the computational tools to process qualitative text data at scale, public consultation is frequently reduced to a "tick-box exercise."

Furthermore, Jiang et al. (2016) demonstrated that failing to properly assess public opinion leads directly to stakeholder misalignment and severe construction delays. From a computer science perspective, there is an urgent need to apply advanced text-mining and semantic clustering architectures to automate the extraction of engineering constraints from informal text.

## 3. Literature Survey & Existing Systems

Recent research highlights the growing adoption of NLP in the construction sector. In a comprehensive review, *"Natural language processing for construction engineering: Artificial intelligence-driven text analysis techniques"* (2024), researchers identified current NLP applications, including:

1. Rule-based algorithms checking architectural blueprints against formal building codes.

2. Named Entity Recognition (NER) applied to formal construction safety accident reports.

3. Information extraction from structured procurement documents.

Furthermore, in a comprehensive review by Waqar et al. (2024) titled *'Natural language processing for construction engineering: Artificial intelligence-driven text analysis techniques'*, the authors mapped out the current state-of-the-art NLP applications across the civil sector. Their findings reinforce the growing necessity for automated text analysis in engineering and validate the shift toward AI-driven data extraction

**Limitations of Existing Systems (The Research Gap):**

Current state-of-the-art models excel at processing *formal, structured* engineering documents. However, they suffer severe accuracy degradation when applied to *informal, noisy, and unstructured public chatter*. There is currently no end-to-end AI pipeline capable of extracting physical sustainability entities from informal citizen feedback, clustering duplicate demands mathematically, and executing a multi-signal priority ranking.

**4. Proposed System**

We propose an automated, multi-stage Artificial Intelligence pipeline designed to bridge this gap. The system consists of four primary modules:

1. **Data Ingestion & Noise Filtering:** A DistilBERT transformer model fine-tuned to perform binary classification, isolating actionable sustainability requirements from general public chatter.

2. **Custom NER Engine:** A spaCy-backed BERT model trained on a custom ontology to extract domain-specific entities (e.g., [STAKEHOLDER], [INFRASTRUCTURE COMPONENT], [SUSTAINABILITY METRIC]).

3. **Semantic Clustering Module:** Utilizing Sentence-Transformers to generate vector embeddings, followed by UMAP for dimensionality reduction and HDBSCAN for density-based clustering to group duplicate complaints.

4. **Prioritization Cascade:** An LLM-audited algorithmic cascade that ranks clustered infrastructure requests based on environmental impact severity, regulatory urgency, and stakeholder consensus.

**5. Hardware & Software Requirements**

**Software Requirements:**

- **Language:** Python 3.10+
- **ML/AI Frameworks:** PyTorch, HuggingFace Transformers, spaCy, scikit-learn
- **Clustering Libraries:** umap-learn, hdbscan, sentence-transformers
- **Dataset:** 1.34 GB EU Public Consultation Dataset (JSON/CSV)

**Hardware Requirements:**

- **Processor:** Intel Core i5/i7 (11th Gen or higher) or AMD Ryzen equivalent.
- **RAM:** Minimum 16 GB (32 GB recommended for vector embedding generation).
- **GPU:** NVIDIA RTX 3060 or higher (CUDA supported) for accelerated transformer training.

## 6. Expected Outcomes

- An end-to-end NLP pipeline capable of processing 10,000+ unstructured public comments in under an hour.
- A structured decision-support dashboard output for civil engineers.
- A high-accuracy classification model explicitly tailored for informal sustainability discourse.

## References


1. Falco, E., Kleinhans, R., & Nunes, C. (2019). *A Theoretical Framework for the Evaluation of Massive Digital Participation Systems in Urban Planning*. Journal of Geovisualization and Spatial Analysis.

2. Jiang, H., Lin, P., & Qiang, M. (2016). *Assessment of online public opinions on large infrastructure projects: A case study of the Three Gorges Project in China*. Environmental Impact Assessment Review.

3. *Natural language processing for construction engineering: Artificial intelligence-driven text analysis techniques*. ScienceDirect (2024).

4. *Integrating Artificial Intelligence into Smart Infrastructure Management for Sustainable Urban Planning*. Technologies (2025).

5. Waqar, A., et al. (2024). *Natural language processing for construction engineering: Artificial intelligence-driven text analysis techniques*. ScienceDirect.